



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL
SENIOR CERTIFICATE

GRADE 12

ENGINEERING GRAPHICS AND DESIGN P2

NOVEMBER 2024

MARKS: 100

TIME: 3 hours

This question paper consists of 6 pages.

Barcode label

DO NOT FOLD THE QUESTION PAPER IN HALF.

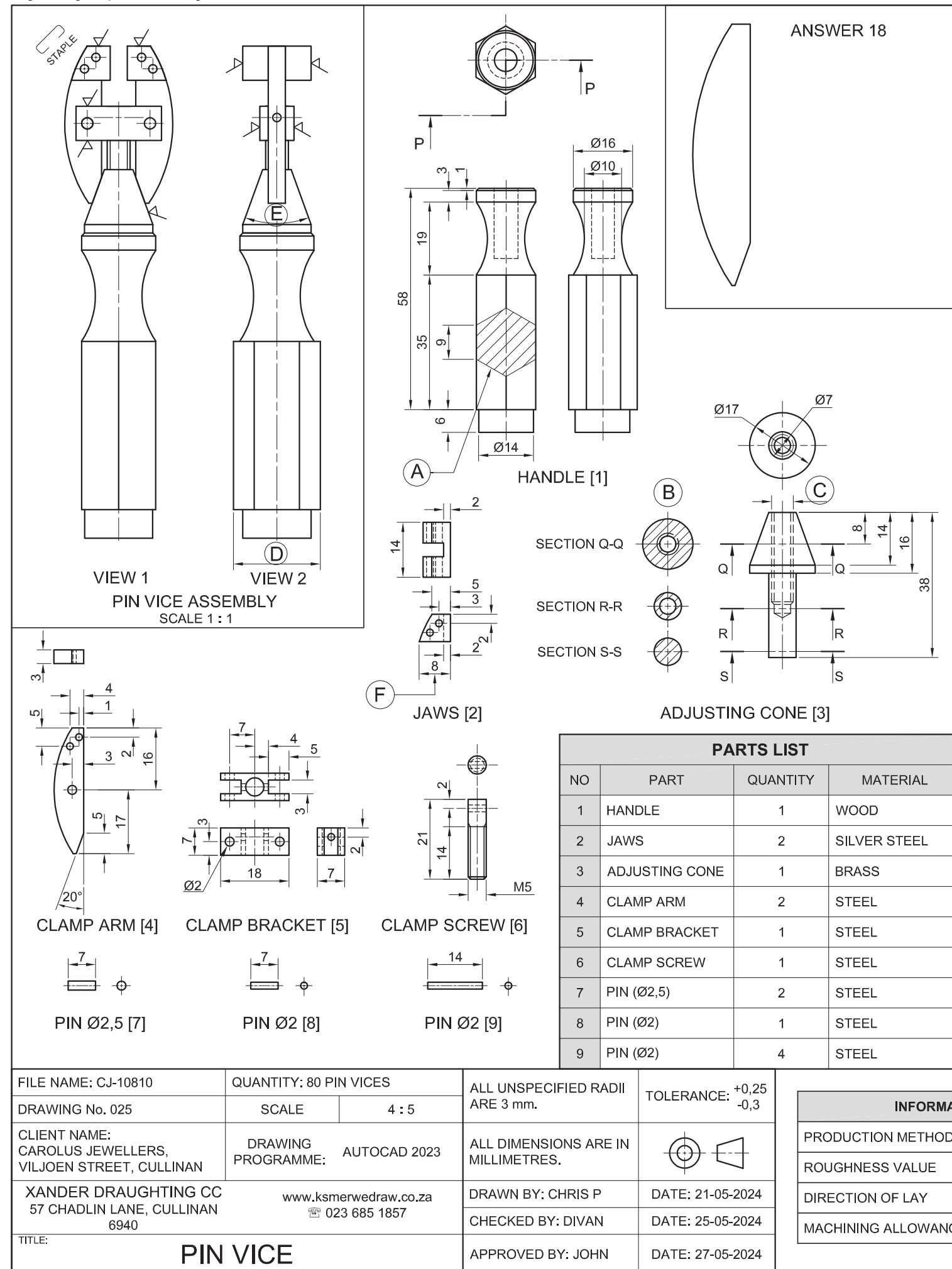
INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.
2. Answer ALL the questions.
3. ALL drawings are in third-angle orthographic projection, unless otherwise stated.
4. ALL drawings must be prepared using pencil and instruments, unless otherwise stated.
5. ALL answers must be drawn accurately and neatly.
6. ALL the questions must be answered on the QUESTION PAPER, as instructed.
7. ALL the pages, irrespective of whether the question was attempted or not, must be re-stapled in numerical sequence in the TOP LEFT-HAND CORNER ONLY.
8. Time management is essential in order to complete all the questions.
9. Print your examination number in the block provided on every page.
10. Any details or dimensions not given must be assumed in good proportion.

FOR OFFICIAL USE ONLY															
QUESTION	MARKS OBTAINED			$\frac{1}{2}$	SIGN	MODERATED			$\frac{1}{2}$	SIGN	RE-MARKING			$\frac{1}{2}$	SIGN
1															
2															
3															
4															
TOTAL															
	2	0	0			2	0	0			2	0	0		

FINAL CONVERTED MARK	CHECKED BY
100	

COMPLETE THE FOLLOWING:
CENTRE NUMBER
CENTRE NUMBER
EXAMINATION NUMBER
EXAMINATION NUMBER



QUESTION 1: ANALYTICAL (MECHANICAL)

Given:

Two views of the assembled parts of a pin vice assembly, detailed views of the individual parts of the assembly drawn to a different scale, a parts list, a title block and a table of questions. The drawings are not presented to the indicated scales.

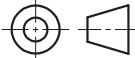
Instructions:

Complete the table below by neatly answering the questions, which refer to the accompanying drawings, title block and mechanical content. [30]

QUESTIONS		ANSWERS		
1	What unit of measurement has been used on the drawing?		1	
2	On what date was the drawing prepared?		1	
3	What is the title of the drawing?		1	
4	What is the telephone number of the draughting company?		1	
5	If VIEW 1 is the front view, what would VIEW 2 be called?		1	
6	What material is required for the handle?		1	
7	What scale is indicated for the individual parts?		1	
8	How many Ø2 pins are there in a single pin vice assembly?		1	
9	How many pin vices must be manufactured?		1	
10	Name the type of section shown at A.		1	
11	Name the type of section shown at B.		1	
12	What type of section would be produced by cutting plane P-P?		1	
13	How many surfaces of the pin vice assembly must be machined?		1	
14	Name the part on which the depth of the threaded hole for the clamp screw (part 6) would be measured.		1	
15	Determine the complete dimensions at: C:	D:	2	
16	Determine the complete angle at E.		1	
17	With reference to the tolerance, determine the maximum and the minimum size for the dimension at F.	MAXIMUM:	1	
		MINIMUM:	1	
18	In the space provided above the views of the adjusting cone (ANSWER 18), use drawing instruments to determine the centre point of the arc. Show ALL construction.		4	
19	In the space below (ANSWER 19), draw, in neat freehand, a machining symbol, with labels for the information and processes provided in the table, for the given surface.		4	
20	In the space below (ANSWER 20), draw, in neat freehand, the <i>SANS 10111</i> conventional representation for a bearing.		3	
TOTAL			30	

ANSWER 19

ANSWER 20

FILE NAME: CJ-10810	QUANTITY: 80 PIN VICES		ALL UNSPECIFIED RADII ARE 3 mm.	TOLERANCE: +0,25 -0,3
DRAWING No. 025	SCALE	4 : 5		
CLIENT NAME: CAROLUS JEWELLERS, VILJOEN STREET, CULLINAN	DRAWING PROGRAMME: AUTOCAD 2023		ALL DIMENSIONS ARE IN MILLIMETRES.	
XANDER DRAUGHTING CC 57 CHADLIN LANE, CULLINAN 6940	www.ksmerwedraw.co.za  023 685 1857		DRAWN BY: CHRIS P	DATE: 21-05-2024
			CHECKED BY: DIVAN	DATE: 25-05-2024
TITLE: PIN VICE			APPROVED BY: JOHN	DATE: 27-05-2024

INFORMATION AND PROCESSES	
PRODUCTION METHOD	GRINDING
ROUGHNESS VALUE	0,5
DIRECTION OF LAY	PERPENDICULAR
MACHINING ALLOWANCE	0,2



SURFACE

EXAMINATION NUMBER	
EXAMINATION NUMBER	2



QUESTION 2: LOCI

NOTE: Answer QUESTIONS 2.1 and 2.2

2.1: MECHANISM

- Given:**
- A schematic drawing of a mechanism consisting of crank OA, connecting rod AB and rocker CD
 - The position of centre point O on the drawing sheet
- Specifications:**
- The positions of centre point O and point C of rocker CD are fixed.
 - Connecting rod AB is pin-jointed to crank OA at A.
 - Rocker CD is pin-jointed to AB at D.

Motion:
As crank OA rotates for one complete rotation, rocker CD oscillates around point C.

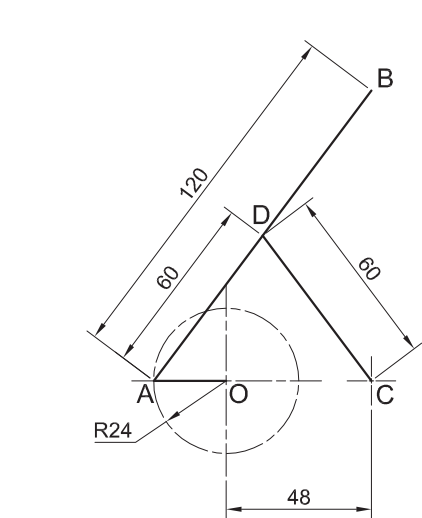
- Instructions:**
- Using centre point O, draw, to scale 1 : 1, the given schematic drawing of the mechanism and label ALL the given points.
 - Trace the locus generated by point B for ONE complete rotation of crank OA.
 - Show ALL construction.

2.2: CAM

Given:
The bottom left-hand corner of the displacement graph, marked P, on the drawing sheet.

- Motion:**
A cam starts at the **maximum** displacement of 64 mm and imparts the following motion:
- There is a dwell period for the first 30°.
 - It descends to its minimum displacement over the next 90° with uniform acceleration and retardation.
 - It rises 36 mm over the next 45° with uniform motion.
 - There is a dwell period for the next 30°.
 - It then descends 22 mm over the next 30° with uniform motion.
 - There is a dwell period for the next 45°.
 - It returns to its original position over the last 90° with simple harmonic motion.

- Instructions:**
- Using corner P as the 0° position, draw, to a rotational scale of 120 mm = 360° and a displacement scale of 1 : 1, the displacement graph for the given motion.
 - Label the displacement graph and include the rotational scale.
 - Show ALL construction.



P
0°

ASSESSMENT CRITERIA 2.1				
1	GIVEN	6 1/2		
2	CONSTRUCTION	5		
3	LOCUS OF B	8		
PENALTIES (-)				
TOTAL		19 1/2		

ASSESSMENT CRITERIA 2.2				
1	GRAPH CONSTRUCTION	6		
2	POINTS + CURVE	11 1/2		
3	LABELS	1		
PENALTIES (-)				
SUBTOTAL 2.2		18 1/2		
SUBTOTAL 2.1		19 1/2		
TOTAL		38		

EXAMINATION NUMBER	
EXAMINATION NUMBER	
3	



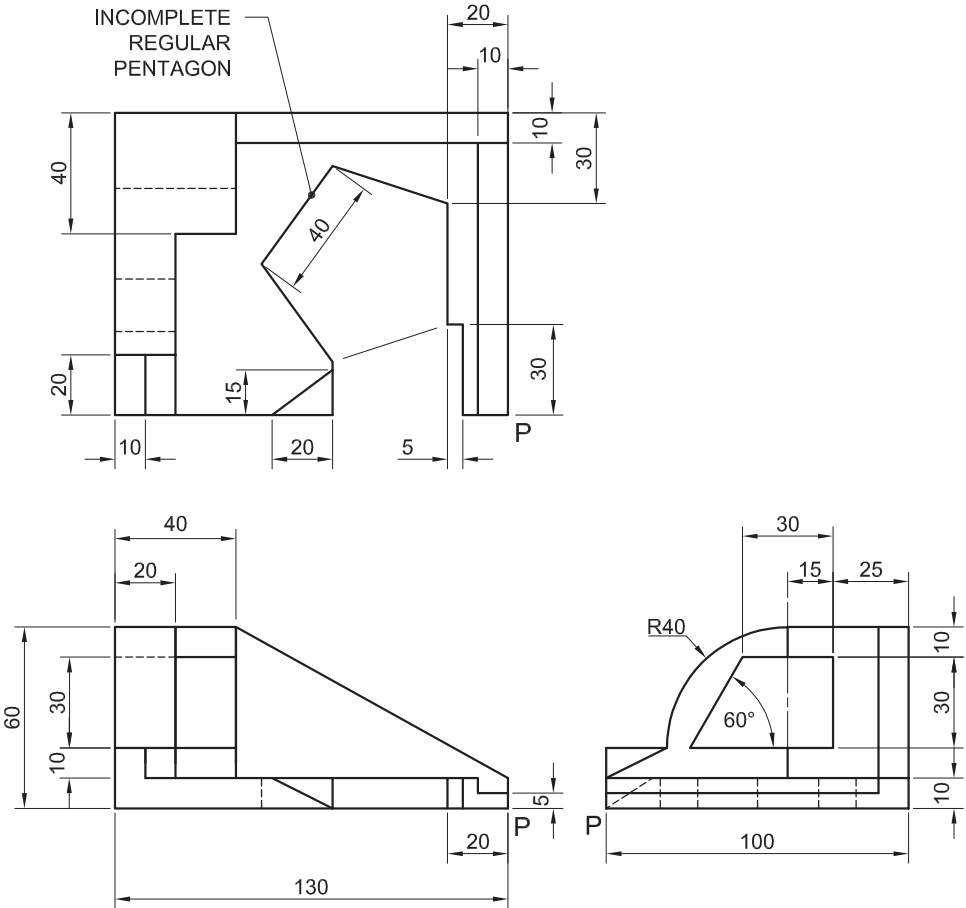
QUESTION 3: ISOMETRIC DRAWING

- Given:**
- The front view, top view and right view of a casting
 - The position of point P on the drawing sheet

Instructions:
Using scale 1 : 1, convert the orthographic views of the casting into an isometric drawing.

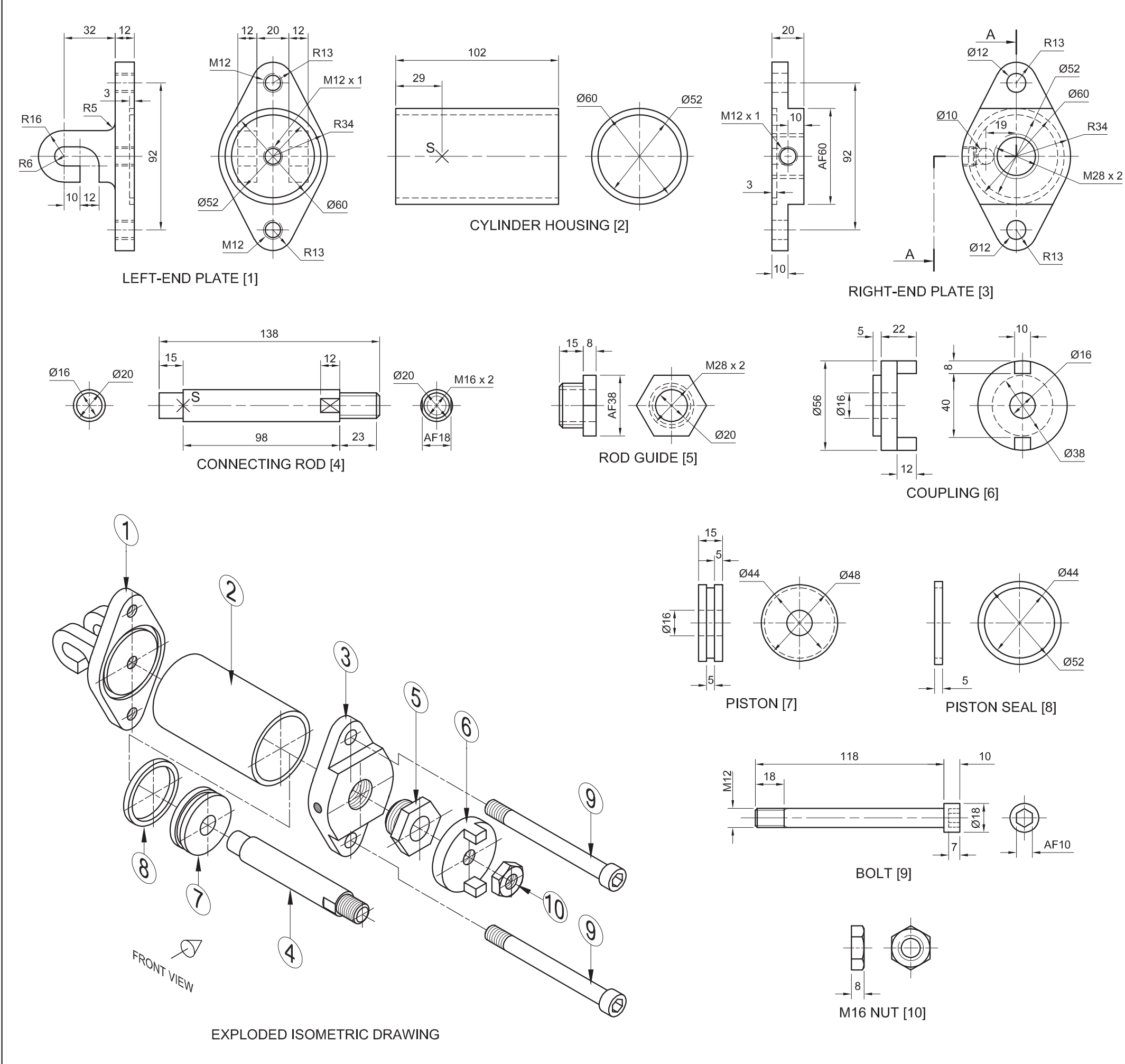
- Use P as the starting and lowest point of the drawing.
- Show ALL construction.
- NO hidden detail is required.

[40]



P

ASSESSMENT CRITERIA					
1	PLACING + AUX. VIEW	2 1/2			
2	BASE + PENTAGON	20			
3	UPPER PORTION	14 1/2			
4	CIRCLES + CONSTR. + CL	3			
PENALTIES (-)					
TOTAL		40			
EXAMINATION NUMBER					
EXAMINATION NUMBER					
					4



QUESTION 4: MECHANICAL ASSEMBLY

- Given:**
- The exploded isometric drawing of the parts of a hydraulic cylinder assembly, showing the position of each part relative to all the others
 - Orthographic views of each of the parts of the hydraulic cylinder assembly

- Instructions:**
- Answer this question on page 6.
 - Draw, to scale 1 : 1 and in third-angle orthographic projection, the following views of the assembled parts of the hydraulic cylinder assembly:
 - 4.1 **The right view**
 - 4.2 **A half-sectional front view** on cutting plane A-A. Show the top half of the assembly in section, as seen from the direction of the arrow on the exploded isometric drawing. The cutting plane is shown on the right view of the right end plate (part 3).

- NOTE:**
- Planning is essential.
 - ALL drawings must comply with the *SANS 10111* guidelines.
 - The convention of symmetry may NOT be applied.
 - Align point S on the connecting rod (part 4) with point S on the cylinder housing (part 2).
 - Show THREE faces of the M16 nut (part 10) on the half-sectional front view.
 - NO hidden detail is required.
- [92]

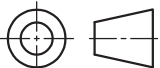
PARTS LIST			
PART		QUANTITY	MATERIAL
1	LEFT-END PLATE	1	STAINLESS STEEL
2	CYLINDER HOUSING	1	MILD STEEL
3	RIGHT-END PLATE	1	CAST IRON
4	CONNECTING ROD	1	STAINLESS STEEL
5	ROD GUIDE	1	BRASS
6	COUPLING	1	CAST IRON
7	PISTON	1	ALUMINIUM ALLOY
8	PISTON SEAL	1	RUBBER
9	BOLT	2	MILD STEEL
10	M16 NUT	1	MILD STEEL

SCGGENGINEERING CC

13 PRESSURE AVE
SOUTH PARK
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092 400 2590

HYDRAULIC CYLINDER

ALL DIMENSIONS ARE IN MILLIMETRES.



5



FOR OFFICIAL USE ONLY	
INCORRECT ORTHOGRAPHIC PROJECTION	
INCORRECT OVERALL SCALE	
INCORRECT HATCHING	
PARTS NOT ASSEMBLED	
TOTAL PENALTIES (-)	

ASSESSMENT CRITERIA					
RIGHT VIEW					
		POSSIBLE	OBTAINED	SIGN	MODERATED
1	RIGHT-END PLATE	5			
2	BOLTS	4			
3	COUPLING	3 ¹ / ₂			
4	M16 NUT	4			
SUBTOTAL		16 ¹ / ₂			
HALF-SECTIONAL FRONT VIEW					
1	LEFT-END PLATE	10 ¹ / ₂			
2	CYLINDER HOUSING	5			
3	RIGHT-END PLATE	9			
4	CONNECTING ROD + PISTON + SEAL	11			
5	ROD GUIDE	3			
6	COUPLING	6			
7	BOLTS	14			
8	M16 NUT	4			
SUBTOTAL		62 ¹ / ₂			
GENERAL					
1	CENTRE LINES	3			
2	ASSEMBLY	10			
SUBTOTAL		13			
TOTAL		92			
PENALTIES (-)					
GRAND TOTAL					
EXAMINATION NUMBER					
EXAMINATION NUMBER					6